

# Context-Aware Cyberbullying Detection using Transformer Model

K. Shanmukha Reddy

Department of Electronics and Communication  
Engineering,  
Amrita School of Engineering,  
Amrita Vishwa Vidyapeetham, Chennai

N. Lokeshwar Varma

Department of Electronics and Communication  
Engineering,  
Amrita School of Engineering,  
Amrita Vishwa Vidyapeetham, Chennai

G. Dyuti

Department of Electronics and Communication  
Engineering,  
Amrita School of Engineering,  
Amrita Vishwa Vidyapeetham, Chennai

**Abstract**—In recent years, cyberbullying is an emerging concern due to the popularity of social media sites and its impact on the mental health and online safety of users. This paper proposes a "Context-Aware Cyberbullying Detection Framework" based on machine learning and transformer algorithms to classify conversations in social media platforms based on cyberbullying types. The system architecture is based on a series of steps for data preprocessing, feature extraction, model building, evaluation, and prediction. Four classification models have been developed and compared in this paper: Support Vector Machine, Logistic Regression, BERT, and the proposed "Context-Aware BERT" model. The datasets used in this paper consist of 196140 comments with four types of cyberbullying: harassment bullying, hate speech bullying, sexual bullying, and not cyberbullying. The "Context-Aware BERT" model is based on taking two comments as input: previous comment and reply comment. The experimental results show that the accuracy of the BERT model is 90.67% for single comment classification, while the accuracy of the Context-Aware BERT model is 84.81%, with better recall for minority classes: sexual bullying (0.90) and hate speech bullying (0.94). The system is deployed as a Streamlit application for real-time cyberbullying prediction with a cybercrime reporting feature.

**Index Terms**—Cyberbullying Detection, Natural Language Processing, Machine Learning, BERT, Text Classification, Social Media Analytics

## I. INTRODUCTION

With the widespread use of social media platforms, cyberbullying has increasingly become a serious social issue, raising great concerns about users' mental health and safety. The study in [1] offers a review of the latest machine learning approaches coupled with natural language processing solutions for identifying cyberbullying and harassment. Support vector machine models were shown to perform well, but there are issues with sarcasm detection and real-time processing. In [2] is proposed a machine learning-based approach for cyberbullying detection in social networks using textual content, which also pointed out some limitations regarding multilingual and contextual

understanding. Similarly, [3] applies traditional machine learning techniques for the classification of cyberbullying, reaching good accuracy but showing dependency on dataset quality and difficulty in detecting implicit cyberbullying. The proposed [4] work suggests an automated cyberbullying detection system using BERT for feature selection, which has an accuracy of 99.12% by analysing aggressive behaviour from the Formspring, Instagram, and MySpace datasets. The rise [5] in the usage of social media platforms has led to an increase in the number of cyberbullying cases. This paper proposes a method to identify harmful content in tweets using NLP(Natural Language Processing) and machine learning algorithms. A total of 39,870 tweets are used to classify various cases of cyberbullying. The proposed system uses supervised machine learning classifiers to identify bullying acts at an early stage. The proposed [6] work aims at early detection of cyberbullying using supervised machine learning with threshold and dual models based on Bag-of-Words and temporal comment features, resulting in a performance improvement of up to 42% using time-aware evaluation on real-world social media datasets.

Hybrid learning models with the use of ensemble learning models have been suggested in attempts to optimize the result of classification. An example of such a model using ensemble learning for cyberbullying and hate speech identification is presented in [7]. An extensive review of existing datasets, technologies, and methodologies for the identification of cyberbullying is conducted in [8], proving the efficiency of deep learning models together with the associated complexities. Classification using Gradient Boosting is considered in [9], Philipo et al., in their study titled "Evaluation of Various Text Classification Methods for Detecting Cyberbullying on Social Media Sites Using Machine Learning and Transformer Models" (2024), focus on analyzing different text classification methods for cyberbullying detection using machine learning and transformer models. Optimized deep learning models in-

roduced in [10] provide high accuracy, but high computational power is needed. A hybrid machine learning model for cyberbullying detection in Bengali language is shown in [11], which has better performance but cannot be generalized for other languages. Cross-platform and analytical studies also give significant contributions to this domain. In [12], a site-agnostic early detection approach has been proposed, which makes the system platform-independent, though multimodal content like images and videos are not considered. Comparative studies in [13] confirm that deep learning models are superior over traditional ones but they indicate issues with biased datasets. [14] In Gong (Doctoral dissertation), the author examines the applicability of different BERT models for cyberbullying comments classification by exploring their effectiveness in text classification tasks and their performance in identifying abusive language on social media platforms. Finally, [15] reviews current machine learning-based cyberbullying detection methods and identifies the major gaps that at present there is a lack of standard datasets, multilingual support, and real-time deployment capabilities.

## II. MAIN CONTRIBUTION

- To create a multi-class model for detecting cyberbullying on social media platforms by classifying text into various categories of bullying.
- To apply machine learning models such as Support Vector Machine (SVM) and Logistic Regression, along with the BERT transformer model, for cyberbullying classification.
- To perform feature extraction using techniques such as TF-IDF, sentence embeddings, and tokenization to convert textual data into numerical representations.
- To implement preprocessing techniques for cleaning noisy textual data by removing unwanted elements such as special characters, URLs, and duplicate entries.
- To develop a system that enables real-time cyberbullying prediction through terminal and Streamlit application interfaces.
- To integrate a reporting feature in the Streamlit application where, if the confidence percentage is greater than 40%, a report button is displayed that redirects users to the official cybercrime reporting website.

## III. LITERATURE SURVEY

The results showed that many Instagram media sessions with a large percentage of negative comments were not cases of cyberbullying, but comments posted in a short time frame were strongly correlated with bullying. The accuracy of the proposed classifiers, Naïve Bayes and Linear SVM, with multimodal features such as text, images, and metadata, reached a maximum of 0.87. The uniqueness of the approach is that it analyses multimodal data to differentiate cyberbullying from cyberaggression, which is not possible with text analysis. The method relies on highly negative sessions and a small number of labelled data, which is a limitation for generalization to other social media platforms [16]. The results of

the performance table showed that the deep learning models greatly enhance the detection of cyberbullying, with BERT having the highest accuracy of 87.3% and CNN-LSTM (Convolution Neural Network-Long Short Term Memory) second with 86.5% accuracy on the Facebook comment dataset. The classifiers used were CNN, LSTM, CNN-LSTM, and BERT. The originality of the work is in the proposed hierarchical CNN-LSTM architecture with contextual embeddings. The shortcomings are in using English text data, not analysing multimodal, not supporting multilingual, and not assessing adversarial robustness [17]. The experiment showed that automatic cyberbullying detection using linear SVM with sentiment, and topic-based features resulted in promising F1 scores of 64.32% for English and 58.72% for Dutch. The originality of the research is in the detection of signals of bullies, victims, and bystanders. The shortcomings are in the difficulty of implicit bullying, irony, misspellings, and contextual victim responses because of the absence of world knowledge [18]. Pratama et al. (2024) have presented a model for multiclass cyberbullying detection by utilizing the BERT model, and its performance is better than that of traditional machine learning methods in classifying text on social media into various categories of bullying. [19]. The results showed that NLP preprocessing with machine learning classifiers enhances the detection of cyberbullying, attaining 94.2% accuracy with the Random Forest classifier on a total of 39,870 Twitter messages. The classifiers Random Forest, SVM (Support Vector Machine), Logistic Regression, Naive Bayes, and KNN (K-Nearest Neighbours) were used. The originality of the research is in the multi-class classification of cyberbullying categories using SMOTE (Synthetic Minority Over-sampling Technique)-balanced data. The cons are in the dependence on text data and the weakness in dealing with contextual subtleties like sarcasm [20]. The experiment revealed that the hybrid CNN-BiLSTM (Convolution Neural Network – Bidirectional Long Short Term Memory) model with stacked GloVe and FastText embeddings resulted in an accuracy of 98% in multilingual cyberbullying detection on real-time Twitter data. Classifiers are CNN, LSTM, BiLSTM. The originality of the research is in the multilingual real-time detection of cyberbullying using hybrid embeddings. However, the research gaps are in the dependence on text data and the absence of bullying detection through images or videos [21]. The experiment revealed that the transformer-based XLMRoBERTa model attained the highest accuracy of 82.61% and F1-score of 0.83 for identifying Bengali cyberbullying content. The classifiers used in this experiment include SVM, RF, MNB(Multinomial Naive Bayes), CNN, GRU(Gated Recurrent Unit), LSTM, BiLSTM, m-BERT(Multi-BERT), and Bangla-BERT. The originality of the research is in the multilingual dataset, which is resource-constrained. However, there are some research gaps [22]. The experiment showed that the BiLSTM model had the best accuracy of 91% for multiclass cyberbullying classification with a dataset of 47,692 tweets. The models used for classification were BiLSTM, CNN-BiLSTM, BiLSTM-GRU. The gaps are in overfitting problems, the use of text-only data,

and the inability to distinguish between non-cyberbullying and cyberbullying samples [23]. The experiment revealed that the hyperparameter-optimized MTF-IDF with the LGBM(Light Gradient Boosting Machine) classifier obtained an accuracy of 83.82% in identifying cyberbullying from 47,692 tweets. The classifiers used were LGBM, XGB(Extreme Gradient Boosting), RFC, DTC, KNC, and BNB. The originality is in the use of entropy-adjusted TF-IDF with Optuna optimization. However, the gaps are in the use of text-based features and the absence of real-time adaptive learning capabilities for dynamic platforms [24]. The proposed MLP-based deep learning model with TF-IDF and unigram hybrid features was found to have an accuracy of 93.2% in Turkish cyberbullying detection on 5000 Twitter posts. The classifiers used were MLP, CNN, LSTM, and BLSTM. The originality of the work is in the combination of text and social media features with data augmentation using synthetic data. The shortcomings are in the computational complexity and adaptability to multiple languages [25]. The experiment revealed that the combination of big data analytics with machine learning and NLP enhances early cyberbullying detection. The classifiers used in the experiment include SVM, Random Forest, CNN, RNN, LSTM, and Decision Trees. The originality of the research is in the combination of behavioral and text features for prediction. The areas that need to be explored include privacy issues, bias, and multimodal fusion [26]. The experiment showed that pre-trained language models with transfer learning can greatly enhance the accuracy of cyberbullying detection, with MiniLM reaching 97.84% crossvalidation accuracy and 98.57% hold-out accuracy on the AMiCA dataset. The models tested include classifiers like DistilBERT, DistilELECTRA. The strength of the approach is in the fine-tuning of PLMs(Product Lifecycle Management) for better understanding of context. The weaknesses include dependence on text-based single-platform corpora and lack of generalization to multilingual communities [27]. The findings showed that cross-platform cyberbullying detection using transfer learning in AI improves generalization and sensitivity in social media sessions. The classification models are rule-based, machine learning, deep learning, and transformer-based PLMs like BERT and RoBERTa. The originality is in the XP-CB(Explainable Prediction for CyberBullying), LS-CB(Lexical Semantic CyberBullying), IDXCB(Implicit Detection of eXtreme CyberBullying), and EAT(Emotion-Aware Transformer) frameworks. However, the research gaps are in swear word bias, session-based models, cross-platform generalization, and indirect cyberbullying detection [28]. The paper concluded that supervised machine learning with NLP methods enhances the detection of cyberbullying by examining the harmful intent, repetition, and abusive language patterns on social media posts. The classifiers used in the paper include SVM, Naïve Bayes, and Logistic Regression. The originality of the paper is in the severity-based categorization and language analysis. The weaknesses of the paper include sarcasm detection, sensitivity to spelling, and lack of contextual understanding [29]. The proposed method resulted in an average accuracy of 80.83% in identifying cyberbullying from 11,101

Twitter comments using Logistic Regression with SMOTE. The proposed method used the Logistic Regression classifier for binary classification. The uniqueness of the proposed method is in using class balancing and cross-validation for stability. The gaps in the proposed method are in the reduction of testing accuracy and the inability to consider contextual sarcasm in text data [30].

## IV. METHODOLOGY

### A. Overview of Proposed System

The system proposed is a "Context-Aware Cyberbullying Detection Framework" that can automatically detect cyberbullying content in social media conversations. The system is different from existing systems as it takes into account both the previous comment and reply comment simultaneously before making a prediction, unlike existing systems that only consider a single comment for prediction.

The system implements and compares four different models: SVM, Logistic Regression, BERT, and the proposed "Context-Aware BERT" model. The system categorizes social media text into four categories: "not\_cyberbullying," "harassment\_bullying," "hate\_speech\_bullying," and "sexual\_bullying." A real-time prediction interface is also created using Streamlit, where a report button is displayed if bullying is detected with more than 40% confidence, directing the user to the official cybercrime reporting website.

### B. Workflow Description

The workflow of the proposed system is composed of a series of structured steps, in total seven steps. In the data collection phase, social media conversation pairs consisting of previous comments and reply comments are collected and labeled with different cyberbullying categories. In the data preprocessing phase, special characters, URLs, duplicate data, and null data are removed from the data set. Additionally, all the words in the data set are converted to lowercase. In the case of the proposed Context-Aware BERT model, each reply comment is paired with the previous comment.

In the feature extraction phase, TF-IDF Vectorization is used for the SVM model, sentence embeddings are used for the Logistic Regression model, and BERT tokenization with attention masks is used for the BERT and Context-Aware BERT models. The data is then split into 70% training data, 15% validation data, and 15% testing data. All four models will be trained and evaluated with the following parameters: Accuracy, Precision, Recall, F1-Score, and Confusion Matrix with class weights.

Finally, the trained models are deployed using a Streamlit application to predict cyberbullying in real-time. If the model is confident about detecting cyberbullying with more than 40%, a report button is displayed, redirecting the user to the official cybercrime reporting website.

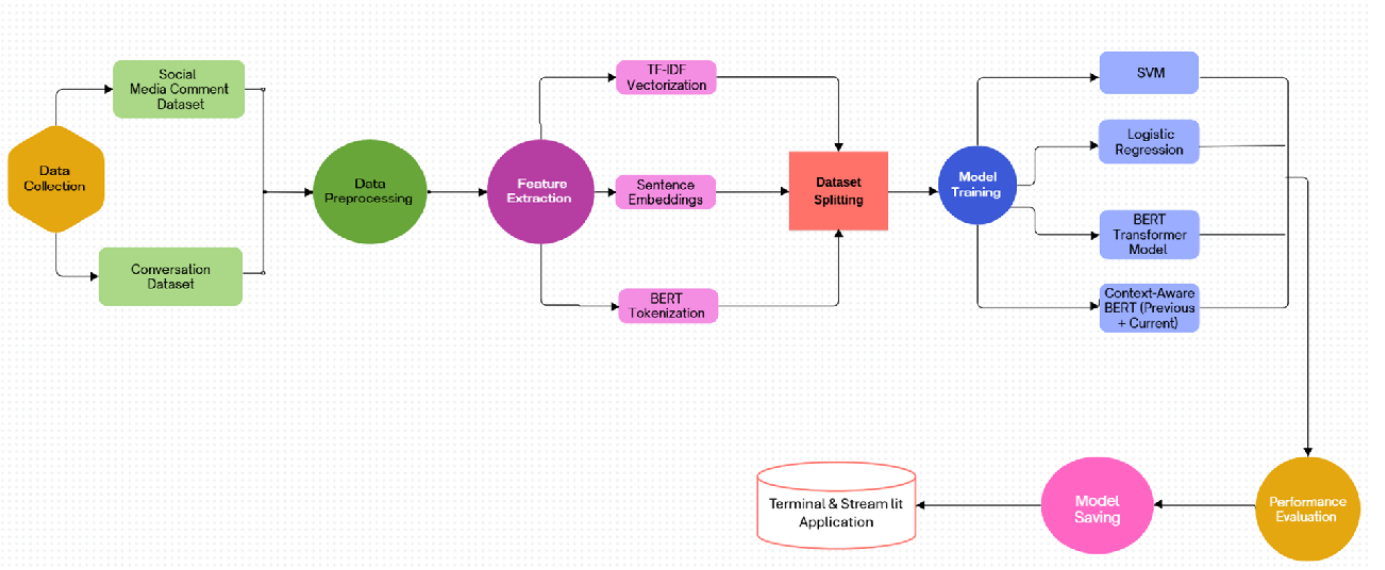


Fig. 1. Block Diagram

### C. Block Diagram

## V. DATA PREPROCESSING STEPS

### A. Introduction to Data Preprocessing

The data collected from the internet may contain noises like URLs, special characters, spacing, and duplicates. It may affect the learning process of the models. Hence, the data needs to be preprocessed before the feature extraction process. The main idea behind data preprocessing is to keep the data in a consistent format while preserving the original meaning.

### B. Text Cleaning

The unwanted data like URLs, special characters, and unnecessary data are removed using the regular expression method.

### C. Text Normalization

All text data is converted to lowercase to ensure consistency with similar words. Example: "You Are Useless!" → "you are useless" Excess spaces normalization is done using Whitespaces normalization, where spaces between words are kept consistent.

### D. Removal of Special Characters

Non-alphanumeric characters and unwanted punctuation marks are eliminated, keeping the text structure intact.

### E. Handling Missing and Duplicate Data

Rows with missing text or labels are dropped to avoid training errors. Duplicate messages are removed to prevent bias and overfitting during learning.

### F. Context Window Construction

This step is specifically added for the Context Aware BERT model. In the baseline models such as SVM, Logistic Regression, and BERT, only the reply comment is used as input. However, it is not clear from the reply comment whether it is a bullying comment or not. For instance, the reply comment itself may look absolutely clean and innocent when it is "yes may or may not." However, when we look at the previous comment that is "will you sleep with me tonight," it is clear that it is a sexual bullying comment. Hence, both the previous comment and reply comment are combined together and fed into the model.

However, for each sample in Dataset 2, the previous comment and reply comment are combined into a single input. The previous comment is first because it is the background information. Then the reply comment is the message being classified. The combined input takes the form: [CLS] previous\_comment [SEP] reply\_comment [SEP]

The combined input is then passed through the BERT tokenizer. The tokenizer turns the text into numbers that the model learns.

## VI. ALGORITHM

### A. Introduction

The proposed system uses a variety of classification algorithms for cyberbullying detection in social media text data. Both traditional machine learning algorithms and deep learning algorithms, also known as transformer algorithms, are considered for the proposed system. The classification algorithms considered for the proposed system are as follows: Support Vector Machine (SVM) Logistic Regression BERT Transformer Model Context-Aware BERT Model

### B. Overall Algorithm Procedure

Loading the datasets Cleaning and preprocessing the text data For baseline models, use only reply comment as input For context-aware BERT model, use previous comment and reply comment as input Transform text into numerical form using TF-IDF, sentence embeddings, or BERT tokenization Split data into training sets, validation sets, and test sets Train all four models Evaluate models using performance metrics Save the trained models Use saved models for predictions to classify whether the input is cyberbullying or not

### C. Support Vector Machine (SVM) Algorithm

Support Vector Machine is a supervised learning algorithm that separates classes using an optimal hyperplane with the maximum margin. Given training samples:  $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$  The optimization goal is:

$$\min \frac{1}{2} \|w\|^2 + C \sum \xi_i \quad (1)$$

where:

- $w$  represents model weights
- $C$  controls regularization
- $\xi_i$  represents classification error

TF-IDF vectorization is applied to convert text into numerical vectors. The TF-IDF score is calculated as:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t) \quad (2)$$

where:

$$\text{TF}(t, d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total terms in document } d} \quad (3)$$

$$\text{IDF}(t) = \log \left( \frac{\text{Total number of documents}}{\text{Number of documents containing term } t} \right) \quad (4)$$

### Step-by-Step Process

- Input preprocessed reply comment dataset.
- Apply TF-IDF vectorization to turn text into numerical vectors.
- Encode class labels numerically and split data into training, validation, and testing sets.
- Train Linear SVC classifier and predict labels for new data.
- Calculate evaluation metrics and save the trained SVM model.

### D. Logistic Regression Algorithm

Logistic Regression performs probabilistic multi-class classification using sentence embedding features. Each reply comment is converted into a 384-dimensional vector using a pre-trained sentence transformer model. The probability estimation is defined as:

$$P(y = i | x) = \frac{e^{w_i \cdot x}}{\sum_j e^{w_j \cdot x}} \quad (5)$$

where  $x$  represents embedding vectors. The loss function minimized during training is cross-entropy loss:

$$L = - \sum y_i \log(\hat{y}_i) \quad (6)$$

where:

- $y_i$  represents the actual label
- $\hat{y}_i$  represents the predicted probability for class  $i$

### Step-by-Step Process

- Generate sentence embeddings from cleaned reply comment text.
- Encode labels using a label encoder and split embeddings into training, validation, and testing sets.
- Train Logistic Regression classifier and optimize weights using gradient learning.
- Compute prediction probabilities and select class with the highest probability.
- Calculate evaluation metrics and save the trained model.

### E. BERT Transformer Algorithm (Baseline)

BERT is a deep learning model based on the transformer architecture. It is able to comprehend the context or relationship between words using self-attention. The text is tokenized into sub-words, and the special tokens [CLS] and [SEP] are added. The context of the token [CLS] is used for classification. The input format is defined as: [CLS] reply\_comment [SEP] The self-attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left( \frac{QK^T}{\sqrt{d_k}} \right) V \quad (7)$$

where:

- $Q$  represents the Query matrix
- $K$  represents the Key matrix
- $V$  represents the Value matrix
- $d_k$  represents the dimension of key vectors

The loss function minimized during training is cross-entropy loss:

$$L = - \sum y_i \log(\hat{y}_i) \quad (8)$$

where:

- $y_i$  represents the actual label
- $\hat{y}_i$  represents the predicted probability for class  $i$

### Step-by-Step Process

- Tokenize reply comment as [CLS] reply\_comment [SEP] using BERT tokenizer.
- Add padding and attention masks to the token sequence.
- Pass token embeddings through transformer encoder layers.
- Extract contextual sentence representation from [CLS] token output.
- Apply classification layer, fine-tune model parameters using backpropagation and output predicted cyberbullying class.

### F. Context-Aware BERT Algorithm

Context-Aware BERT is a modified version of the transformer-based model that can identify cyberbullying by analyzing the conversation rather than just one comment. Unlike the baseline BERT, which uses only the reply comment, the Context-Aware BERT uses both the previous comment and the reply comment. This enables the model to understand the

relationship between two comments and thus make a more accurate prediction. The contextual representation of the [CLS] token is used for classification. The input format is defined as: [CLS] previous\_comment [SEP] reply\_comment [SEP] The context representation is obtained as:

$$h_{[CLS]} = \text{BERT}([CLS] \text{ prev } [SEP] \text{ reply } [SEP]) \quad (9)$$

The classification output is defined as:

$$\hat{y} = \text{softmax}(W \cdot h_{[CLS]} + b) \quad (10)$$

Where:

- $h_{[CLS]}$  = context representation vector from BERT encoder
- $W$  = weight matrix of classification layer
- $b$  = bias term

The weighted cross-entropy loss minimized during training is:

$$L = - \sum_{i=1}^k w_i \cdot y_i \log(\hat{y}_i) \quad (11)$$

Where:

- $w_i$  = class weight for class  $i$
- $y_i$  = actual label
- $\hat{y}_i$  = predicted probability for class  $i$

Step-by-Step Process:

- Previous comment and reply comment will be combined in the form of [CLS] previous\_comment [SEP] reply\_comment [SEP]. Tokenization will be done using the BERT tokenizer.
- Padding and attention mask will be added to the tokenized combined comment.
- 12 transformer encoder layers will be used to get token embeddings.
- CLS token output will be used to get context representation  $h_{[CLS]}$ .
- Classification will be done using a classification layer.
- Backpropagation will be used to fine-tune model parameters using weighted cross-entropy.
- Cyberbullying class will be predicted with a confidence score.

#### G. Prediction Algorithm

In the process of deployment, the following steps are used for prediction:

- User input text is received.
- Preprocessing steps are performed.
- The text is converted into feature form.
- The trained model is loaded.
- Prediction is generated.
- The label is decoded.
- Cyberbullying category is displayed through the terminal or Streamlit app.

TABLE I  
TOOLS AND LIBRARIES USED

| Tool / Library           | Purpose                                   |
|--------------------------|-------------------------------------------|
| Python                   | Primary programming language              |
| PyTorch                  | Deep learning framework for BERT training |
| HuggingFace Transformers | BERT model and tokenizer                  |
| Scikit-learn             | SVM and Logistic Regression models        |
| Sentence Transformers    | Sentence embedding generation             |
| Pandas and NumPy         | Data preprocessing and manipulation       |
| Matplotlib               | Performance comparison graphs             |
| Streamlit                | Real-time prediction interface            |

TABLE II  
HARDWARE AND SOFTWARE ENVIRONMENT

| Component               | Details           |
|-------------------------|-------------------|
| Operating System        | Windows           |
| Programming Language    | Python 3.14.3     |
| Deep Learning Framework | PyTorch           |
| BERT Model              | bert-base-uncased |
| Training Device         | GPU               |

## VII. RESULT AND EVALUATION

### A. Experimental Setup

Tools and Libraries Used:

Hardware and Software Environment:

Dataset Details: In this study, there are two datasets. The first one is the classification dataset with a large number of single comments. This classification dataset has about 1,96,140 comments. This classification dataset is used to train the SVM, logistic regression, and BERT baseline models.

The second one is the previous comment and reply comment pairs with the label of bullying. The size of the data is 1,96,140. This data is used to train the context-aware BERT model.

The data size of 1,96,140 is divided into three sections. Out of the data size of 1,96,140, 1,37,344 is used for training. 29,431 is used for validation and remaining 29,431 is used for testing.

### B. Evaluation Results

The dataset is divided into training, validation, and test sets, and four classification models are developed. The objective is to compare traditional machine learning models with transformer models for cyberbullying detection.

1) *Support Vector Machine (SVM)*: The SVM model used TF-IDF feature extraction with a linear kernel. The confusion

TABLE III  
SVM VALIDATION AND TEST PERFORMANCE

| Set        | Accuracy | Precision | Recall | F1-score |
|------------|----------|-----------|--------|----------|
| Validation | 81.15%   | 81.36%    | 81.15% | 81.21%   |
| Test       | 80.92%   | 81.06%    | 80.92% | 80.95%   |

matrix indicates strong performance for the not\_cyberbullying class while showing some confusion between harassment and sexual bullying categories.

2) *Logistic Regression*: Logistic Regression was trained using sentence embeddings from pre-trained models. The

TABLE IV  
LOGISTIC REGRESSION VALIDATION AND TEST PERFORMANCE

| Set        | Accuracy | Precision | Recall | F1-score |
|------------|----------|-----------|--------|----------|
| Validation | 80.05%   | 78.98%    | 80.05% | 79.17%   |
| Test       | 80.27%   | 79.31%    | 80.27% | 79.42%   |

model shows balanced performance but faces difficulty in understanding complex bullying language.

3) *BERT Transformer Model*: The BERT model achieved better performance than traditional machine learning models because of its ability to understand contextual meaning of words. The confusion matrix confirms better prediction sta-

TABLE V  
BERT VALIDATION AND TEST PERFORMANCE

| Set        | Accuracy | Precision | Recall | F1-score |
|------------|----------|-----------|--------|----------|
| Validation | 90.56%   | 90.68%    | 90.56% | 90.61%   |
| Test       | 90.67%   | 90.69%    | 90.67% | 90.67%   |

bility across all cyberbullying categories compared to SVM and Logistic Regression.

4) *Context-Aware BERT Model*: The Context-Aware BERT model was trained using paired conversation inputs — previous comment and reply comment combined as [CLS] previous comment [SEP] reply comment [SEP].

TABLE VI  
CONTEXT-AWARE BERT VALIDATION AND TEST PERFORMANCE

| Set        | Accuracy | Precision | Recall | F1-score |
|------------|----------|-----------|--------|----------|
| Validation | 84.23%   | 87.35%    | 84.23% | 84.82%   |
| Test       | 84.81%   | 87.78%    | 84.81% | 85.36%   |

TABLE VII  
CONTEXT-AWARE BERT DETAILED PER-CLASS PERFORMANCE (TEST SET)

| Class                | Precision | Recall | F1-score | Support |
|----------------------|-----------|--------|----------|---------|
| harassment_bullying  | 0.66      | 0.88   | 0.76     | 4559    |
| hate_speech_bullying | 0.80      | 0.94   | 0.85     | 4642    |
| not_cyberbullying    | 0.97      | 0.81   | 0.88     | 18690   |
| sexual_bullying      | 0.61      | 0.90   | 0.72     | 1541    |
| Weighted avg         | 0.88      | 0.85   | 0.85     | 29432   |

The Context-Aware BERT model shows improved recall for minority classes especially sexual bullying (0.90) and hate-speech bullying (0.94) by using conversational context during classification.

#### C. Experiment 1 — Baseline Comparison Table

All four models are compared on the test set using Accuracy, Precision, Recall, and F1-score. BERT has the best

TABLE VIII  
COMPARISON OF ALL MODELS ON TEST SET

| Model               | Accuracy | Precision | Recall | F1-score |
|---------------------|----------|-----------|--------|----------|
| Logistic Regression | 80.27%   | 79.31%    | 80.27% | 79.42%   |
| SVM                 | 80.92%   | 81.06%    | 80.92% | 80.95%   |
| BERT (Baseline)     | 90.67%   | 90.69%    | 90.67% | 90.67%   |
| Context-Aware BERT  | 84.81%   | 87.78%    | 84.81% | 85.36%   |

overall accuracy of 90.67% on a single comment classification problem. The Context-Aware BERT model has an accuracy of 84.81% on a conversation-based dataset, which is a more difficult problem as it requires understanding the relationship between two comments. Context-Aware BERT has a better recall rate for minority classes such as sexual\_bullying at 0.90 and hate\_speech\_bullying at 0.94, which are the most important classes to detect in a cyberbullying detection system.

#### D. Experiment 2 — Context Ablation Study

The ablation study indicates how much improvement is achieved in terms of classification performance due to the context of the conversation. Two models are compared: BERT without context, and BERT with one previous comment as context. Adding one previous comment as context enhances

TABLE IX  
CONTEXT ABLATION STUDY

| Model Configuration            | Accuracy | Precision | Recall | F1-score |
|--------------------------------|----------|-----------|--------|----------|
| BERT (no context — reply only) | 90.67%   | 90.69%    | 90.67% | 90.67%   |
| BERT + 1 previous comment      | 84.81%   | 87.78%    | 84.81% | 85.36%   |

the recall for minority classes such as sexual bullying and hate speech bullying. The baseline model’s incorrect classification of some comments as not\_cyberbullying is corrected when the previous comment is included as context. This demonstrates that context in conversations allows the model to grasp the underlying meaning behind apparently innocuous short replies that may be misinterpreted in isolation.

#### E. Experiment 3 - Error Analysis

From the result of the confusion matrices, it can be seen that most of the wrong predictions occur between the classes of "not cyberbullying" and "harassment bullying" because, without any context, the reply comment of both classes is almost similar. The SVM and Logistic Regression models cannot classify the classes of "not cyberbullying" and "harassment bullying" correctly because the models only consider word frequency, but not the meaning of the conversation. Although the performance of the Baseline BERT model is better than the performance of the SVM and Logistic Regression models, it also cannot predict the class of cyberbullying correctly because, without any context, the reply comment may not contain any offensive words. However, with the use of the Context-Aware BERT model, the wrong predictions are minimized because the model can read both the previous comment and



the reply comment, thus understanding the meaning of the conversation correctly.

#### F. Performance Comparison Graph

The performance comparison graphs are shown in Fig. 2, Fig. 3, Fig. 4

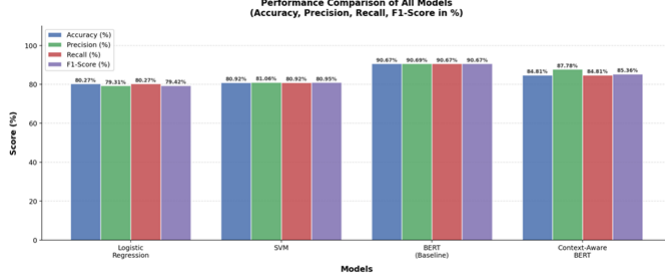


Fig. 2. Performance Comparison Graph

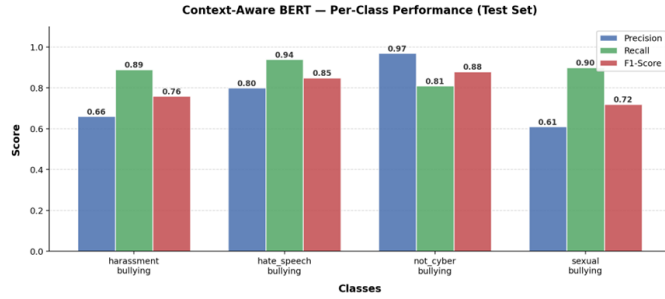


Fig. 3. Context-Aware BERT - per class performance Graph

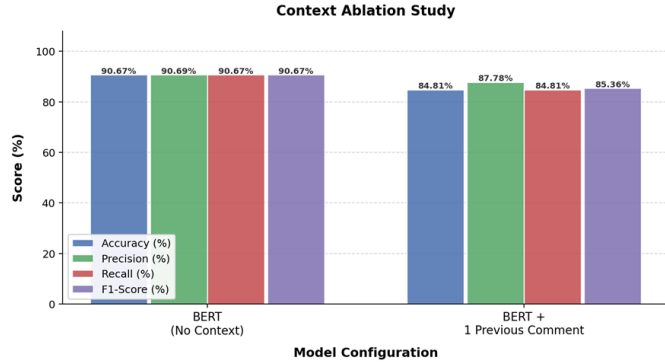


Fig. 4. Context Ablation Study Graph

#### G. Discussion

SVM and Logistic Regression achieve 80% accuracy due to their reliance only on word frequency, failing to account for deeper text meaning. The baseline model, BERT, achieves the best accuracy of 90.67%. This is because it understands the real meaning of words through self-attention mechanisms. Context-Aware BERT achieves an accuracy of 84.81%, which is on a more difficult conversation-based dataset. It also

has a much better recall value for minority classes such as sexual\_bullying (0.90) and hate\_speech\_bullying (0.94). The similarity between validation accuracy (84.23%) and test accuracy (84.81%) indicates that this model is not overfitting.

#### H. Comparison with Existing Works

The proposed BERT baseline achieves 90.67% accuracy, which is higher than most of the existing traditional models and deep learning models. The Context-Aware BERT achieves 85.08% accuracy on conversation-level classification, which is a more difficult classification task compared to single comment classification, which is used by most of the existing models. Most of the existing models do not address the minority class classification, while the proposed model achieves 0.90 recall for sexual bullying and 0.94 recall for hate speech bullying.

### VIII. CONCLUSION

This study proposes a framework called "Context-Aware Cyberbullying Detection Framework" that uses machine learning and transformer models to detect cyberbullying in social media conversations. Two datasets are used: a baseline dataset containing around 196,000 single comments and a conversation-based dataset containing 196,140 pairs. The dataset contains four classes: harassment bullying, hate speech bullying, sexual bullying, and not cyberbullying. Four models are implemented: SVM, Logistic Regression, BERT, and Context-Aware BERT. SVM achieves 80.92% accuracy, and Logistic Regression achieves 80.27% accuracy. The baseline BERT model achieves 90.67% accuracy with an F1-score of 0.91. The proposed Context-Aware BERT model achieves 84.81% accuracy and improved recall for minority classes: sexual\_bullying (0.90) and hate\_speech\_bullying (0.94), by analyzing both previous and reply comments using "[CLS] previous\_comment [SEP] reply\_comment [SEP]".

This is further supported by error analysis, which shows that most misclassifications occur between not\_cyberbullying and harassment\_bullying because many harmful conversations cannot be identified from a single reply alone. When the previous comment is included, the model correctly classifies conversations that may appear harmless in isolation but are actually bullying. The close match between validation accuracy (84.23%) and test accuracy (84.81%) indicates that the model is not overfitting and confirms the effectiveness of transformer-based context-aware cyberbullying detection for intelligent content moderation tools.

### REFERENCES

- [1] Islam MS, Orno AN, Arifuzzaman M. Approach to social media cyberbullying and harassment detection using advanced machine learning. Available at SSRN 4705261. 2024 Mar 12.
- [2] Syafiq AM, Ab Razak MF, Abidin AF, Mohamad S, Kamaruddin NK. Social network approach for cyberbullying detection using machine learning. Journal of Governance and Integrity. 2025 Jul 30;8(1):874-80.
- [3] Kagi S. Cyberbullying detection using machine learning. Journal of Scientific Research and Technology. 2025 Nov 22:148-57.
- [4] Mali MK, Pawar RR, Shinde SA, Kale SD, Mulik SV, Jagtap AA, Tambewagh PA, Rajput PU. Automatic detection of cyberbullying behaviour on social media using Stacked Bi-Gru attention with BERT model. Expert Systems with Applications. 2025 Mar 1;262:125641.



- [5] Sayed FR, Elnashar EH, Omara FA. Cyberbullying detection in social media using natural language processing. *Scientific African*. 2025 Jun 1;28:e02713.
- [6] Lopez-Vizcaíno MF, Novoa FJ, Carneiro V, Cacheda F. Early detection of cyberbullying on social media networks. *Future Generation Computer Systems*. 2021 May 1;118:219-29.
- [7] Mubeen M, Muskan A, Akram A, Rashid J, Alshalali TA, Sarwar N. Cyberbullying-Related Automated Hate Speech Detection on Social Media Platforms Using Stack Ensemble Classification Method. *International Journal of Computational Intelligence Systems*. 2025 Jul 9;18(1):174.
- [8] Philipo AG, Sarwatt DS, Ding J, Daneshmand M, Ning H. Cyberbullying detection: Exploring datasets, technologies, and approaches on social media platforms. *ACM Computing Surveys*. 2024.
- [9] Gaston Philipo A, Sarwatt DS, Ding J, Daneshmand M, Ning H. Assessing Text Classification Methods for Cyberbullying Detection on Social Media Platforms. *arXiv e-prints*. 2024 Dec:arXiv-2412.
- [10] Neelakandan S, Sridevi M, Murugeswari K, Sridevi R. Deep learning approaches for cyberbullying detection and classification on social media. *Computational Intelligence and Neuroscience*. 2022 Jun 11;2022:2163458.
- [11] Akhter A, Acharjee UK, Talukder MA, Islam MM, Uddin MA. A robust hybrid machine learning model for Bengali cyber bullying detection in social media. *Natural Language Processing Journal*. 2023 Sep 1;4:100027.
- [12] Lopez-Vizcaíno M, Novoa FJ, Artières T, Cacheda F. Site agnostic approach to early detection of cyberbullying on social media networks. *Sensors*. 2023 May 16;23(10):4788.
- [13] Unnava S, Parasana SR. A study of cyberbullying detection and classification techniques: A machine learning approach. *Engineering, Technology & Applied Science Research*. 2024 Aug 2;14(4):15607-13.
- [14] Gong SM. Text classification of cyberbullying comments: a study on the applicability of various BERT models (Doctoral dissertation, ETSI Informatica).
- [15] Sultan D, Omarov B, Kozhamkulova Z, Kazbekova G, Alimzhanova L, Dautbayeva A, Zholdassov Y, Abdrakhmanov R. A Review of Machine Learning Techniques in Cyberbullying Detection. *Computers, Materials & Continua*. 2023 Mar 1;74(3).
- [16] Hosseinmardi H, Mattson SA, Rafiq RI, Han R, Lv Q, Mishra S. Detection of cyberbullying incidents on the instagram social network. *arXiv preprint arXiv:1503.03909*. 2015 Mar 12.
- [17] Khafajeh H. Cyberbullying Detection in Social Networks Using Deep Learning. *International Arab Journal of Information Technology (IAJIT)*. 2024 Nov 1;21(6).
- [18] Van Hee C, Jacobs G, Emmery C, Desmet B, Lefever E, Verhoeven B, De Pauw G, Daelemans W, Hoste V. Automatic detection of cyberbullying in social media text. *PloS One*. 2018 Oct 8;13(10):e0203794.
- [19] Pratama MY, Sarno R, Septiyanto AF, Haryono AT, Sabilla SI. Multi-class Cyberbullying Detection Using BERT. In *2024 Beyond Technology Summit on Informatics International Conference (BTS-I2C) 2024 Dec 19 (pp. 498-502)*. IEEE.
- [20] Sayed FR, Elnashar EH, Omara FA. Cyberbullying detection in social media using natural language processing. *Scientific African*. 2025 Jun 1;28:e02713.
- [21] Raj M, Singh S, Solanki K, Selvanambi R. An application to detect cyberbullying using machine learning and deep learning techniques. *SN Computer Science*. 2022 Jul 26;3(5):401.
- [22] Sihab-Us-Sakib S, Rahman MR, Forhad MS, Aziz MA. Cyberbullying detection of resource constrained language from social media using transformer-based approach. *Natural Language Processing Journal*. 2024 Dec 1;9:100104.
- [23] Jaradat G, Shehab M, Ibrahim D, Najdawi S, Sihwail R. Deep learning approaches for detecting cyberbullying on social media. *Journal of Computational and Cognitive Engineering*. 2025 Mar 31;4(4):451-65.
- [24] Vivekananth P, Sharma N. Detecting Cyberbullying in Social Media: An NLP-Based Classification Framework. *Indian Journal of Science and Technology*. 2025 Feb 18;18(5):380-9.
- [25] Aliyeva C, O, Yaganoğlu M. Deep learning approach to detect cyberbullying on twitter. *Multimedia Tools and Applications*. 2025 Jun;84(19):20497-520.
- [26] Pranith BY. Machine Learning Solutions for Cyberbullying Detection and Prevention on Social Media.
- [27] Bong JK, Varathan KD, Teng TH. Cyberbullying Detection in Social Media Using Pre-Trained Language Models. *Malaysian Journal of Computer Science*. 2025 Mar 30;38(1):29-54.
- [28] Yi P. Cyberbullying Detection Across Different Social Media Platforms (Doctoral dissertation, Queen Mary, University of London).
- [29] Perera A, Fernando P. Cyberbullying detection system on social media using supervised machine learning. *Procedia Computer Science*. 2024 Jan 1;239:506-16.
- [30] Prayudani S, Adha LT, Ariyani T, Lubis AR. Prediction of Cyberbullying in Social Media on Twitter Using Logistic Regression. *Journal of Applied Informatics and Computing*. 2025 Aug 7;9(4):1681-6.